SAMPLE

Cosmos Unveiled: The Cosmic Neutrino Background

Echoes of the Big Bang

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Free sample: Introduction & Chapter 1

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Introduction

You are being watched, right now, by the oldest light in the universe. Not the light of stars, which is young by cosmological standards, and not the microwave glow of the big bang's afterglow, which is merely old, but something older and stranger: a sea of neutrinos, the universe's first matter, set loose in the first second of everything, and flowing through you, and through the Earth, and through every wall ever built, at a density of hundreds per cubic centimeter, continuously, since the beginning. This book is about that sea, the cosmic neutrino background: what it is, where it came from, why it has never been directly detected, and why cosmologists consider it one of the most important objects in the sky that no one has ever seen.

The neutrino is physics' most antisocial particle, a particle that interacts so weakly with everything else that a neutrino produced in the sun's core will usually fly out of the sun, across a hundred million kilometers of space, through the Earth, and keep going, without ever touching anything. In the first second of the big bang, when the universe was dense and hot beyond imagining, even neutrinos collided, and the universe's first matter soup was in thermal contact with itself, neutrinos scattering off electrons and positrons and photons like birds in a flock. Then the universe expanded, the flock thinned, and at about one second of age the neutrinos stopped colliding with anything, decoupled, and streamed free, a background that has been coasting, cooling, and thinning ever since, unchanged except in temperature, for thirteen point eight billion years.

The parallel with the famous microwave background is exact and worth savoring. The photons of the big bang's glow were set free a bit later, at three hundred and eighty thousand years, when the universe cooled enough for atoms to form and the light stopped scattering, and that glow, detected in the 1960s, is now the most precisely measured object in all of science. The neutrinos were set free earlier, at one second, and they carry information the photons cannot: a picture of the universe at an age the light does not remember, the universe as a one-second-old, nuclear-density plasma, older than any atom, older than any nucleus of any atom in your body. If we could see the neutrino background, we would see the universe's true baby photo.

The catch, the reason this book is about an unseen sea, is that seeing it is almost impossible, and the book will be honest about the difficulty. The relic neutrinos are cold, slow, and feebler than any particle ever detected, and the experiments proposed to catch them require heroic measures, tons of tritium, years of integration, techniques not yet built. But the sea is not invisible in every sense: it has been detected indirectly, in the chemistry of the first

nuclei, in the pattern of the galaxy distribution, in the fine details of the microwave glow itself, and the book's middle chapters are the story of how a particle nobody has caught has been measured anyway, by its footprints.

The chapters ahead run from the particle to the sea to the hunt:

A note on the book's cousins, because a series owes anyone the map. The microwave glow has its own volume in this series, and the two books are siblings, telling the story of the two afterglows from their two sides, the light's and the ghost's, and anyone who takes both will notice the deliberate handoffs: this book treats the glow only as the sea's thermometer's neighbor, and the glow's book returns the favor. The dark matter and the structure

formation have their volumes too, and the follower who continues there will find the neutrino's fingerprints already in hand, the mass bound and the small-scale suppression, a piece of the sky's census, carried forward, which is the series' method: each book stands alone, and the books together build the one cosmology, the reader holding the parts.

The neutrino's own strange biography, the first second's physics, the background's present-day character, its fingerprints in the sky, its role in the structure of the cosmos, the connections to dark matter, the direct-detection proposals, and the open questions, ending where the subject ends, at the frontier. The tone is the series' standing promise: real physics, honestly labeled, with the speculation marked, and the wonder, as always, in the facts.

One last note before the chapters, on the book's honest boundaries. The neutrino background is the coldest, oldest subject in this series' cosmology shelf, and it touches, in the literature, a dozen neighboring fields, the particle theory, the dark matter, the early universe, and anyone who wants those fields' full accounts will find them in their own volumes, and this book will hand off at the borders, the series' standing practice. What the book claims to own is the sea itself: the particle's biography, the first second's manufacture, the present-day portrait, the indirect evidence, the hunt, and the open files, told at reading length, in the series' voice, with the same standards the reader has met before, and anyone who finishes it will know the sea as the field knows it, a confirmed relic, a measured ghost, and a portrait waiting, which is, the field would say, the honest state of the art.

Chapter 01: The Ghost in the Equations: Predicting the Background

The cosmic neutrino background has a curious distinction among the universe's great discoveries: it was predicted, in essentially its modern form, decades before anyone had confirmed the neutrino itself existed, and the prediction, once made, sat in the literature like an unopened letter, waiting for

the rest of physics to catch up.

The story begins with the particle. The neutrino was proposed in 1930 as a bookkeeping fix, a desperate theoretical suggestion to balance a nuclear reaction's accounts: beta decay, the process in which a neutron turns into a proton and spits out an electron, seemed to violate the conservation of energy, the electron carrying away less than the books required, and rather than abandon conservation, a young theorist proposed that a new particle, invisible, nearly massless, and undetectable by any instrument of the era, carried away the missing energy under the table. The physics community received the suggestion with a skepticism the proposal's own author shared, and the particle was not found, conclusively, until a quarter century later,

when a reactor's flood of neutrinos, streaming out of the fission cores, was caught by an experiment that put a tank of detector fluid next to a reactor and waited, catching, by the theorists' own arithmetic, a handful of interactions a day from a flux that would pass through light-years of lead.

Once the particle was real, cosmology's arithmetic seized on it almost immediately.

The mid-century's prediction deserves its full context, because the neutrino background was not predicted alone. The hot big bang theory, in the same era's papers, was shown to have made a photon sea as well, an afterglow of the early plasma, and the photon prediction, made on the same pages by the same group of theorists, was vindicated spectacularly when the microwave glow was found in the 1960s, and the theory's stock rose accordingly, and the account should make plain the neutrino prediction's position in the family of predictions: it was the third leg of a stool whose other two legs, the glow and the light elements, were confirmed, and the neutrino sea's turn, the field knew, would come when the techniques caught up, a confidence the decades have begun to repay.

The big bang, in the mid-century's emerging theory, was a machine for making particles: a hot dense plasma, cooling, would produce every species the temperature allowed, in thermal equilibrium, in known numbers, and the neutrino, the lightest of the matter particles, would be made in enormous abundance in the first second, when the universe was hot enough for neutrino pair production to keep pace with the universe's expansion. The theorists of the 1960s, working out the hot big bang's census, realized the neutrinos could not simply disappear: the same weak interaction that made them would stop making them as the universe cooled, and the neutrinos, once made, would be too weakly interacting to ever be destroyed or absorbed, so the early universe's neutrino sea would survive to the present, cooled and diluted by the

expansion, but still there, still passing through everything, the universe's oldest relic, predicted with pencil and paper before anyone imagined how to look.

The prediction's structure takes a paragraph, because it is a preview of the book. The background's temperature, the theorists calculated, would be a couple of degrees above absolute zero, slightly cooler than the microwave background, for a reason the later chapters explain. Its density would be hundreds of particles per cubic centimeter, everywhere, in you, in the Earth, in the space between the galaxies. And its detectability, the same calculation admitted, would be vanishingly small: a neutrino of that temperature, hitting the best detector imaginable, would interact essentially never, and the

prediction, the theorists wrote in the era's honest style, was a cosmological assertion more than an experimental invitation.

The prediction's history has one more turn anyone should enjoy:

The popularization deserves its note, because the sea entered the culture through a famous slim book. The era's theorists wrote the first-second story for a general audience, a book the field still quotes, whose pages walked the reader through the universe's first minutes with the sea's decoupling as a centerpiece, and a generation of physicists, the field's own historians record, arrived at cosmology through that book's account of the ghost sea, the prediction's cultural life outlasting its undetectability. Anyone can keep the loop this book closes: the account written for the public fifty years ago is the same story the coming decade's instruments may finally confirm, the ghost's biography reaching its portrait chapter, with the same numbers, only the error bars changed.

For decades the cosmic neutrino background was the cosmologist's party trick, quoted as evidence of the big bang's completeness, the theory that had already predicted the microwave glow and the light-element abundances, and the background, undetectable, was the third leg of the stool, invoked and unfalsifiable. The book's later chapters describe how that changed: the sea's indirect effects, in the nuclei, the galaxies, and the microwave glow, were measured one by one, until the background became not an unopened letter but a working scientific object, a measurement's target, its properties constrained by data, and its direct detection, the hunt's chapters describe, is

now an engineering problem rather than a fantasy. The ghost, in other words, has been promoted from rumor to census respondent,

The promotion's instruments get their listing, because anyone should know the sea's witnesses by name and trade. The light-element abundances, measured

in the oldest stars and the pristine gas, the spectroscopists' trade. The microwave glow's power spectrum, mapped by satellite and balloon, the cosmologists'. The galaxy surveys' small-scale counts, the observers' deepest catalogs, and the lensing maps that weigh the structure, and each witness, the chapters show, measures the sea by a different property, the density, the species count, the mass, and the agreement of the witnesses, the field's standing point, is the sea's evidence in the scientific sense: not one measurement but a web of them, each holding the others up, the way a bridge's cables are each held by the rest.

Anyone can carry the image through the book: the ghost has no single witness, and needs none. And the book is its biography. The pencil-and-paper

patience gets a line. The prediction waited a quarter century for its particle, the hypothesized ghost having been proposed before anyone could test it, the account keeping the patience's precedent, the first particle predicted from bookkeeping rather than observation, the ghost file closing with the standing method, the subject began as an arithmetic problem, and the arithmetic was right. The beta-budget scandal gets a line. The prediction came from missing money, the nuclear accounts having refused to balance until a share of the energy was assigned to an escapee, the account keeping the scandal's formality, the hypothesis having been offered as an act of bookkeeping rather than faith, the ghost file finishing with the standing audit, the particle was invented to save arithmetic, and the arithmetic it saved was the century's most careful.

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